

Python + DSA

1. Strings – Consecutive Vowels

- Problem: Find the maximum number of consecutive vowels in a given string.
- Technique Used: Sliding Window / Single Pass
- Logic: Increase count for vowel, reset count for consonant.
- Time Complexity: $O(n)$
- Space Complexity: $O(1)$

2. Strings – RGB Signal Light Problem

- Problem: Predict final signal colors after K minutes.
- Transition Cycle: $R \rightarrow G \rightarrow B \rightarrow R$ (cycle length = 3).
- Optimization: Use $K \% 3$ to avoid large iterations.
- Time Complexity: $O(n)$

3. Strings – Maximum Distance of 'a' and Minimum Distance of 'b'

- X = Maximum distance between any two 'a' characters.
- Y = Minimum distance between any two 'b' characters.
- Approach: Track indices of 'a' and 'b'.
- $X = \text{last_a_index} - \text{first_a_index}$.
- Y = minimum difference between consecutive b indices.
- Final Answer = $X + Y$.